2026 Notebook

Daniel Jang

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The purpose of this yearly notebook is to document large scale progress in projects, any significant mistakes, important derivations, nice tricks that I think will serve me into the future, etc.

September

New Methods

M-estimation, Z-estimation, sandwich variance

Influence Functions, Efficient Influnece Functions, connections between IF/EIF and estimating functions

Mistakes I made

October

Debiased Machine Learning (BIP Project)

Debiased / Double Machine Learning (DML) is a semiparametric estimation method that came out of the economics literature that allows us to obtain an unbiased and efficient estimate for a low-dimensional target parameter ψ (perhaps to do statistical inference) while accounting for high-dimensional nuisance parameters η. However, they have become useful in biostatistics. We need to look no further than one of the papers that Jinbo made us present. I will try to give a high level explaination of the concept, since the theory behind why this works is beyond my understanding for now.

I will restrict our discussion on DML to partially linear regression models, since it will connect nicely to concepts we learned this semester. Let A be a binary treatment whose effect we are interested in, let Y be our continuous outcome (we can extend this beyond continuous outcomes), and let $Z = (Z_1, ... Z_p)$ be a p-dimensional vector of covariates that are not of primary scientific interest. Finally, let $f : Z \to \mathbb{R}$ I think? Might be wrong, where $f \in L_2$. This is to guarantee finite second moments, which then implies finite variance Need more detail here probably... as a detail, L_2 is infinite dimensional... an informal linear algebra proof involving polynomials

The partially linear regression model is given as:

$$Y_i = \beta_0 + \beta_A A_i + f(Z) + \varepsilon$$

Recall the **Frisch-Waugh-lovell Theorem**: For a linear regression model $\mathbb{E}(Y \mid X) = \beta_0 + \boxed{\beta_k} X_K + X_{-k}^\top \beta_{-k} + ...$, where X_{-k} denotes all predictors other than X_k,

$$Y^* = \boxed{\beta_k} X_k^* + \varepsilon^*$$

where $Y^* = Y - \hat{\mathbb{E}}(Y \mid X_{-k})$ is the outcome model, $X_k^* = X_k - \hat{\mathbb{E}}(X_k \mid X_{-k})$ is the variable model, and ε^* is proportional to the residual-maker matrix for the remaining predictors (details are not important for the error term).

We can use the same theorem to find the residuals for the outcome and the variable of interest in the partially linear regression model. It is clear with a partially linear regression model that the outcome model is $Y_{resid} = Y - \hat{\mathbb{E}}[Y \mid f(Z)]$ and the treatment model is $A_{resid} = A - \hat{\mathbb{E}}[A \mid f(Z)]$ (oftentimes, A is denoted as D in the literature). This is where the name *double* comes from, not from this method being automatically doubly robust.

Now the daunting question: How do we estimate obtain $\hat{\mathbb{E}}[Y \mid f(Z)]$ and $\hat{\mathbb{E}}[A \mid f(Z)]$? Well, we use the other part of the method's name: *machine learning* to find the $\hat{f}$ that works. I definitely do not have all the details down yet. I am pretty confident the functional forms here will be different

Thought: What might go wrong with just fitting a machine learning model on the original partially linear regression model? I think it has to do with the functional form of f being the same, the doubleness is not present here

This is all great, but machine learning models are a black box. How do we confirm we obtained an unbiased estimate of our target parameter(s)?: **Neyman Orthogonality**. There is some detail that I don't know if I can go into in time, but trust that the literature has shown asymptotic properties hold

Some references:

1) The canonical paper: Double/Debiased Machine Learning for Treatment and Structural Parameters1 Victor Chernozhukov†, Denis Chetverikov‡, Mert Demirer† Esther Duflo†, Christian Hansen§, Whitney Newey†, James Robins
2) A simpler introduction: https://arxiv.org/abs/2504.08324

I will motivate it using RCT data: https://bingkaiwang.github.io/RCT_Bench-main/website/. Unless I am using a noncollapsible model, adjusting for all covariates as my nuisance and using penalized machine learning should still help in semiparametric efficiency gains compared to non-adjustment or simple adjustment (not entirely convinced by that last point)